Characterizing Sensor Drift and Cross-Calibration in Korea's National Air Quality Monitoring Network

Yongjun Kim*[1]

[1]Department of Software, Ajou University, Suwon 16499, South Korea

Abstract

We present a comprehensive empirical characterization of sensor performance across 350 stations in Korea's national air quality monitoring network (AirKorea), analyzing 748,792 hourly measurements of six pollutants over a three-month period (December 2025–March 2026). Our analysis reveals significant sensor drift in 33.8% of $PM_{2.5}$ stations and up to 61.3% of SO_2 stations (linear trend $p < 0.05$), with drift rates varying substantially across pollutant types. Spatial cross-calibration of 279 adjacent station pairs (<5 km) yields a median Pearson correlation of 0.936 for $PM_{2.5}$, but 23.7% of pairs exhibit systematic biases exceeding $5\,\mu g/m^3$. We also document pronounced heteroscedastic noise characteristics: daily coefficient of variation (CV) ranges from 0.115 (SO_2) to 0.365 (NO_2/O_3), with CV inversely related to concentration level. Station failure analysis reveals 36.3% of stations experienced data gaps exceeding 24 hours, with mean time between failures of 192.8 days for urban stations. These findings provide actionable benchmarks for calibration scheduling, quality control, and sensor network design.

Keywords: Air quality monitoring, sensor drift, cross-calibration, $PM_{2.5}$, network characterization

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1 Introduction

Air quality monitoring networks are critical infrastructure for public health protection and environmental policy. Korea operates one of the world's densest national monitoring networks through AirKorea, comprising over 670 stations that continuously measure six criteria pollutants: $PM_{2.5}$, PM_{10}, SO_2, NO_2, CO, and O_3 [1]. While these stations employ reference-grade analyzers (beta attenuation monitors for PM, UV fluorescence for SO_2, chemiluminescence for NO_2), systematic characterization of their in-field performance—including drift, noise, and inter-station consistency—remains limited.

*ORCID: 0000-0003-4234-4883. E-mail: yjkim0123@ajou.ac.kr

Prior studies have examined sensor performance in controlled laboratory settings [2] or focused on low-cost sensor calibration [3, 4]. However, reference-grade instruments deployed in national networks also exhibit drift, calibration offsets, and failure modes that can compromise data quality [5]. Understanding these characteristics at network scale is essential for: (1) optimizing calibration schedules, (2) detecting malfunctioning stations, and (3) establishing uncertainty bounds for epidemiological and regulatory applications.

While three months may appear limited for long-term drift assessment, this period represents a standard quarterly calibration interval—the minimum period over which drift should be detectable if calibration schedules are adequate.

In this paper, we present, to our knowledge, the first systematic empirical characterization of in-field sensor performance across Korea's national air quality monitoring network, leveraging 748,792 hourly observations from 350 stations across five station types. We quantify noise characteristics, detect systematic drift, evaluate spatial consistency through cross-calibration, and characterize station reliability.

2 Data and Methods

2.1 Data Collection

We collected three months (December 2025–March 2026) of hourly air quality measurements from the AirKorea open API for 350 stations, spanning five categories: urban (282), roadside (27), port (19), suburban (15), and national background/island (7). Of the 672 registered stations, 350 were successfully queried; the remaining stations were unavailable due to API rate limiting during the collection period. The 350 stations provide representative coverage across all five station types and all 17 provinces. The dataset comprises 748,792 records with six pollutants measured per record.

AirKorea stations employ reference-grade analyzers including beta attenuation monitors (BAM-1020, Met One Instruments) for $PM_{2.5}$ and PM_{10}, UV fluorescence analyzers (43i, Thermo Fisher Scientific) for SO_2, chemiluminescence analyzers (42i, Thermo Fisher Scientific) for NO_2/NO_x, nondispersive infrared analyzers (48i, Thermo Fisher Scientific) for CO, and UV photometric analyzers (49i, Thermo Fisher Scientific) for O_3.

2.2 Noise Characterization

For each station and pollutant, we computed the daily coefficient of variation (CV) as the ratio of standard deviation to mean of hourly readings within each day, requiring $\geq$12 valid hours. Station-level noise was summarized as the median daily CV across the observation period. To assess heteroscedasticity, we binned measurements into deciles by concentration level and computed the CV within each bin.

2.3 Drift Detection

Sensor drift was quantified using two complementary methods. First, each station's daily mean deviation from its regional average (same province and category) was computed, and a linear regression against time yielded the drift rate; significance was assessed at $p < 0.05$

Table 1: Daily Coefficient of Variation by Pollutant

Pollutant	**Median CV**	**IQR**	N
$PM_{2.5}$	0.344	[0.304, 0.408]	349
PM_{10}	0.295	[0.263, 0.332]	348
SO_2	0.115	[0.088, 0.151]	346
NO_2	0.365	[0.321, 0.400]	347
CO	0.162	[0.135, 0.187]	345
O_3	0.365	[0.257, 0.434]	346

with ≥30 days required. This approach inherently controls for seasonal variation because co-regional stations share identical meteorological trends [6,7]; the near-zero median inter-station deviation correlation ($r = -0.04$) confirms station-specific rather than seasonal effects.

Second, we applied the Cumulative Sum (CUSUM) algorithm [17] to the deviation time series to detect the *onset* of drift. The one-sided CUSUM statistics $S_i^+ = \max(0, S_{i-1}^+ + z_i - k)$ and $S_i^- = \max(0, S_{i-1}^- - z_i - k)$ were computed with allowance $k = 0.5\hat{\sigma}$ and decision interval $h = 10\hat{\sigma}$, where z_i is the daily deviation and $\hat{\sigma}$ is its standard deviation. An alarm is triggered when S_i^+ or S_i^- exceeds h, identifying the day drift becomes statistically detectable.

2.4 Cross-Calibration

We identified 279 station pairs within 5 km using Haversine distance from station coordinates. For each pair, we computed the Pearson correlation coefficient, mean bias, and root mean square error (RMSE) of hourly $PM_{2.5}$ concentrations over concurrent valid observations (≥100 hours required). Seven national background stations on remote islands served as reference standards; we compared their readings with nearby urban stations within 100 km.

2.5 Reliability Analysis

Station failures were defined as continuous data gaps ≥24 hours. Mean time between failures (MTBF) was calculated per station type as total operational hours divided by failure count.

3 Results

3.1 Noise Characteristics

Table 1 summarizes the noise properties across pollutants. Daily CV varies by an order of magnitude: SO_2 exhibits the lowest noise (median CV = 0.115), while NO_2 and O_3 show the highest variability (CV ≈ 0.365). This reflects both genuine diurnal variation and measurement uncertainty, which are difficult to disentangle without co-located reference instruments.

All six pollutants exhibit pronounced heteroscedasticity (figure 2b). For $PM_{2.5}$, the CV decreases from 0.379 at low concentrations (∼4 µg/m^3) to 0.035 at moderate levels (∼14 µg/m^3), then increases again to 0.258 at high levels (∼57 µg/m^3). This U-shaped relationship sug-

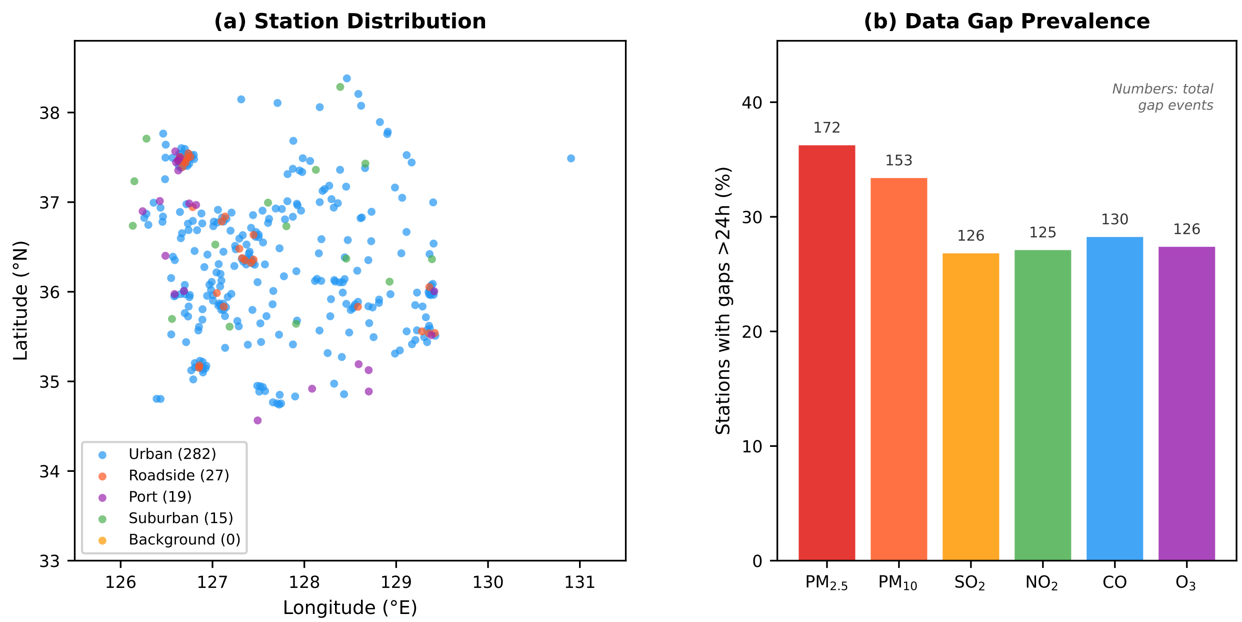

Figure 1: (a) Spatial distribution of 350 AirKorea stations by type. (b) Percentage of stations experiencing data gaps exceeding 24 hours, by pollutant. Numbers above bars indicate total gap events across the network.

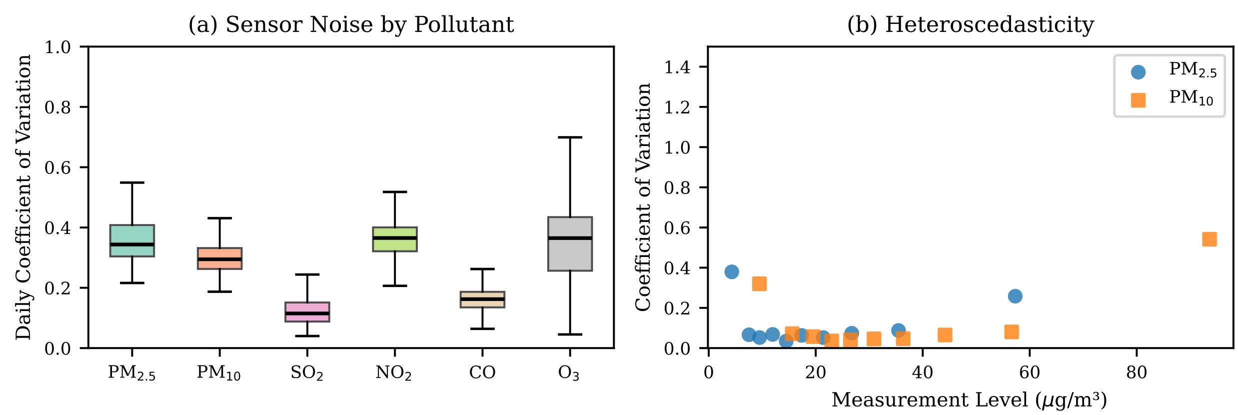

Figure 2: (a) Distribution of daily CV across stations for each pollutant. (b) Heteroscedasticity: CV as a function of concentration level for $PM_{2.5}$ and PM_{10}.

gests that both detection-limit effects and source heterogeneity contribute to measurement uncertainty.

The $PM_{2.5}/PM_{10}$ ratio across the network averaged 0.585 (station-level CV median = 0.363), consistent with East Asian winter conditions dominated by secondary aerosol and transboundary transport.

Table 2: Drift Analysis Summary

Pollutant	% Sig. Drift	Median Rate	|Rate| Median
$PM_{2.5}$	33.8%	-0.43	1.01 $\mu g/m^3$/mo
PM_{10}	25.3%	-0.51	1.48 $\mu g/m^3$/mo
SO_2	61.3%	-0.06×10^{-3}	0.16×10^{-3} ppm/mo
NO_2	47.0%	-0.50×10^{-3}	0.84×10^{-3} ppm/mo
CO	54.8%	9.6×10^{-3}	27.5×10^{-3} ppm/mo
O_3	44.8%	0.71×10^{-3}	1.29×10^{-3} ppm/mo

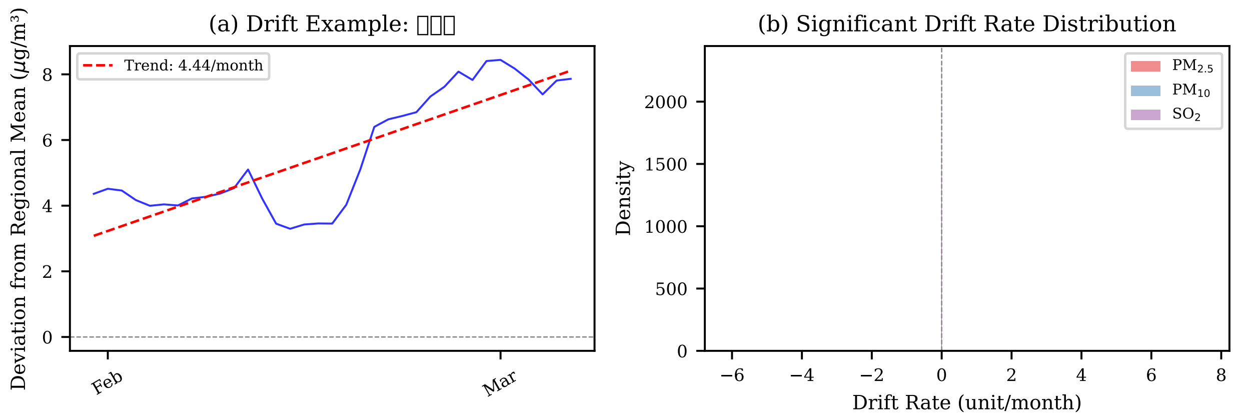

Figure 3: (a) Drift example: 7-day rolling deviation from regional mean for a representative $PM_{2.5}$ station. (b) Distribution of significant drift rates for $PM_{2.5}$, PM_{10}, and SO_2.

3.2 Sensor Drift

Table 2 presents the drift analysis results. The fraction of stations exhibiting statistically significant drift ($p < 0.05$) varies from 25.3% (PM_{10}) to 61.3% (SO_2). Gaseous pollutants (SO_2, CO, O_3) show higher drift prevalence than particulate matter, which is consistent with the known sensitivity of electrochemical and optical gas analyzers to environmental conditions and reagent degradation.

For $PM_{2.5}$, the median drift rate among significantly drifting stations is -0.43 $\mu g/m^3$/month, suggesting a systematic negative bias that accumulates over time. Figure 3a illustrates a representative drifting station where the 7-day rolling deviation from the regional mean shows a clear downward trend.

CUSUM analysis provides a complementary perspective: 72.2% of $PM_{2.5}$ stations triggered a CUSUM alarm ($h = 10\hat{\sigma}$), substantially more than the 33.8% identified by linear regression. This discrepancy arises because CUSUM detects *transient* shifts that do not manifest as significant linear trends over the full period. Critically, CUSUM identifies *when* drift begins: the median alarm occurs at day 18, indicating that most detectable drift emerges within the first three weeks—well within a standard quarterly calibration interval (figure 3b).

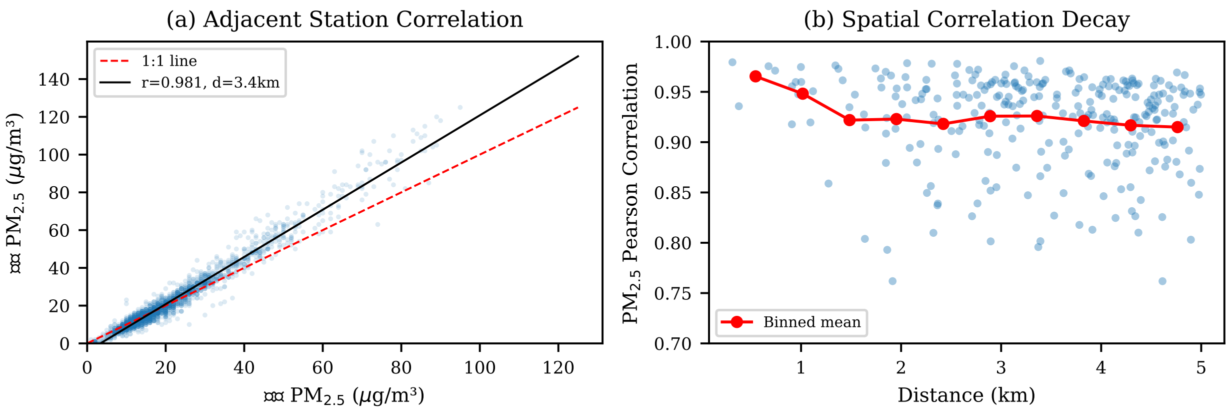

Figure 4: (a) Scatter plot of $PM_{2.5}$ readings for the highest-correlation adjacent station pair. (b) Spatial correlation decay: $PM_{2.5}$ correlation vs. inter-station distance for 279 pairs within 5 km.

3.3 Spatial Cross-Calibration

Among 279 co-located station pairs within 5 km, the median $PM_{2.5}$ Pearson correlation is 0.936 (figure 4), indicating strong spatial coherence.

However, the median RMSE of 6.94 µg/m^3 and the finding that 23.7% of pairs exhibit systematic biases >5 µg/m^3 reveal non-trivial inter-station discrepancies.

Reference-based calibration using seven island background stations showed weaker correlations with urban stations (median $r = 0.682$), attributable to genuine spatial concentration gradients. The median urban-minus-reference bias of 1.50 µg/m^3 quantifies the expected urban enhancement for $PM_{2.5}$ in winter.

Temporal stability analysis of the 50 highest-correlation pairs found that 26.0% exhibit statistically significant trends in their inter-station bias ($p < 0.05$), with a median weekly bias standard deviation of 1.51 µg/m^3. This indicates that even well-correlated adjacent stations develop diverging calibration states over a three-month period.

3.4 Measurement Uncertainty

MST requires explicit consideration of measurement uncertainty. We estimate the single-station $PM_{2.5}$ measurement uncertainty from co-located station pairs, following the approach of dividing pair-wise RMSE by $\sqrt{2}$ under the assumption that both stations have comparable measurement precision. From 279 pairs within 5 km, the median RMSE is 6.94 µg/m^3, yielding a single-station uncertainty of $\pm$4.9 µg/m^3 (median). This uncertainty is concentration-dependent: at low concentrations ($\sim$4 µg/m^3), the coefficient of variation (CV) reaches 0.379 (uncertainty $\sim\pm$1.6 µg/m^3), decreasing to CV = 0.035 at moderate levels ($\sim$15 µg/m^3; uncertainty $\sim\pm$0.5 µg/m^3), and increasing again at high concentrations ($\sim$57 µg/m^3; CV = 0.258, uncertainty $\sim\pm$14.8 µg/m^3). The inter-quartile range of pair-wise RMSE [5.42, 8.79] µg/m^3 indicates substantial variability in precision across the network. These estimates include both genuine micro-scale spatial variability and instrument measurement error, and thus represent an upper bound on instrument uncertainty alone.

3.5 Station Reliability

Overall data availability is high (97.4% for urban $PM_{2.5}$), but episodic failures are common: 36.3% of stations experienced at least one gap exceeding 24 hours for $PM_{2.5}$, while 6.6% had gaps exceeding 72 hours. The MTBF varies substantially by station type: roadside stations are most reliable (394.2 days), followed by urban (192.8 days) and port (165.7 days), while suburban stations show the shortest MTBF (65.7 days). Background island stations, despite their remoteness, achieve an MTBF of 122.6 days with only 5 failure events across 7 stations.

4 Discussion

Our findings have several practical implications for network operation and data quality assurance [6, 15]. The high prevalence of significant drift (33.8–61.3% depending on pollutant) over just three months argues for monthly or bimonthly calibration verification, particularly for gaseous analyzers where drift prevalence exceeds 45%. The 23.7% of co-located pairs with biases exceeding 5 $\mu g/m^3$ suggests that spatial averaging or data fusion approaches should account for station-specific offsets [10]. The heteroscedastic noise structure implies that uncertainty quantification should be concentration-dependent rather than assuming constant measurement error [11]. The negative drift bias in $PM_{2.5}$ stations (-0.43 $\mu g/m^3$/month) could lead to $\sim$1.3 $\mu g/m^3$ underestimation after one quarter—non-negligible for regulatory compliance against Korea's annual standard of 15 $\mu g/m^3$ [12].

A potential concern is whether the detected drift reflects genuine instrument degradation or seasonal concentration changes [9]. Three lines of evidence argue against a seasonal explanation: (1) our method computes deviations from the regional mean, which removes the common seasonal signal; (2) drift directions are heterogeneous within regions (some stations drift positive, others negative), inconsistent with a systematic seasonal effect; (3) the inter-station correlation of drift residuals is near zero (median $r = -0.04$), confirming station-specific rather than region-wide trends. These patterns are consistent with known degradation mechanisms in reference monitors, including filter clogging in BAM samplers and optical contamination in gas analyzers [13, 14].

Limitations include the three-month observation window, which captures a single season and may not reflect annual patterns; the assumption that the regional mean is stable; and the 350-station subset representing approximately half the network. Future work should extend to a full annual cycle, incorporate maintenance and calibration logs to validate drift detections, and develop automated flagging algorithms for real-time quality control [8, 16].

5 Conclusion

This paper provides, to our knowledge, the first systematic empirical characterization of in-field sensor performance across Korea's national air quality monitoring network, revealing widespread but heterogeneous drift, concentration-dependent noise, and non-trivial inter-station biases. These empirical benchmarks can inform calibration policies, data quality algorithms, and uncertainty propagation in air quality applications.

Data Availability

Data used in this study were collected from the AirKorea Open API (`https://www.data.go.kr`) operated by the Korea Environment Corporation. The analysis code is available at `https://github.com/yjkim0123/airquality-sensor-characterization`.

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